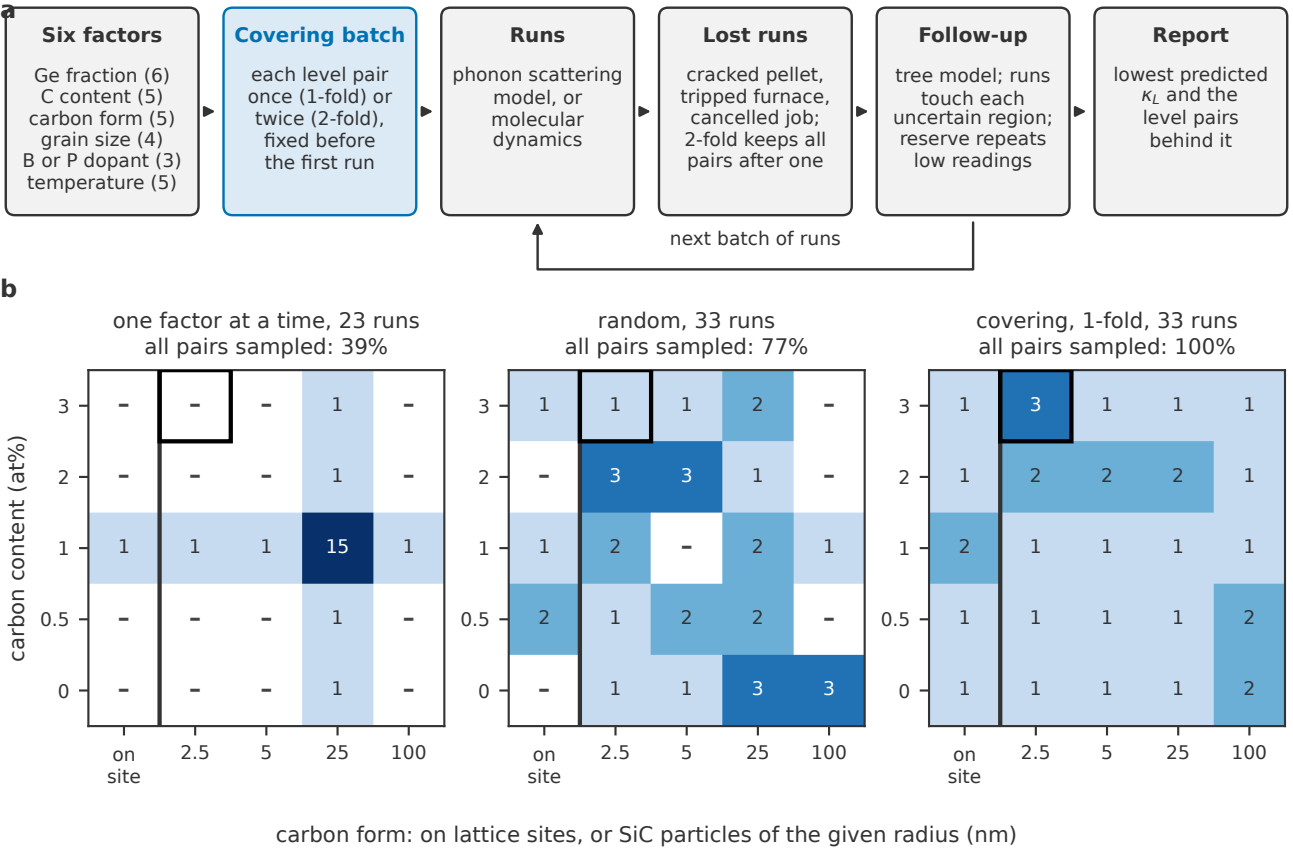
a
Six factors
Ge fraction (6)
C content (5)
carbon form (5)
grain size (4)
B or P dopant (3)
temperature (5)
Covering batch
each level pair once (1-fold) or twice (2-fold), fixed before the first run
Runs
phonon scattering model, or molecular dynamics
Lost runs
cracked pellet, tripped furnace, cancelled job; 2-fold keeps all pairs after one
Follow-up
tree model; runs touch each uncertain region; reserve repeats low readings
Report
lowest predicted κ_L and the level pairs behind it
next batch of runs
b
one factor at a time, 23 runs
all pairs sampled: 39%
random, 33 runs
all pairs sampled: 77%
covering, 1-fold, 33 runs
all pairs sampled: 100%
carbon content (at%)
3
2
1
0.5
0
on site
2.5
5
25
100
carbon form: on lattice sites, or SiC particles of the given radius (nm)